ASSIGNMENT 2: Systematic Design of Web Applications with UML

Course: Web Engineering

Project Title: FocusForge (AI-Enhanced Habit & Mood Tracking Application)

Methodology: UML-based Web Engineering (UWE)

Team: HabbitMasters

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1. Introduction

The objective of this assignment is to systematically design the architecture of our web application, **FocusForge** , using the UML-based Web Engineering (UWE) methodology developed by Rolf Hennicker and Nora Koch.

FocusForge is a cloud-based application that integrates mood tracking with habit monitoring. It applies AI techniques to analyze patterns between emotional states and task completion, providing personalized micro-goals. To model this complex, data-driven web application, we applied the UWE step-by-step process: starting from a domain-specific Conceptual Model, defining navigability in the Navigational Space Model, detailing access mechanisms in the Navigational Structure Model, and finally sketching the UI architecture via the Static Presentation Model.

2. Phase 1: Conceptual Model (Content Model)

Description:

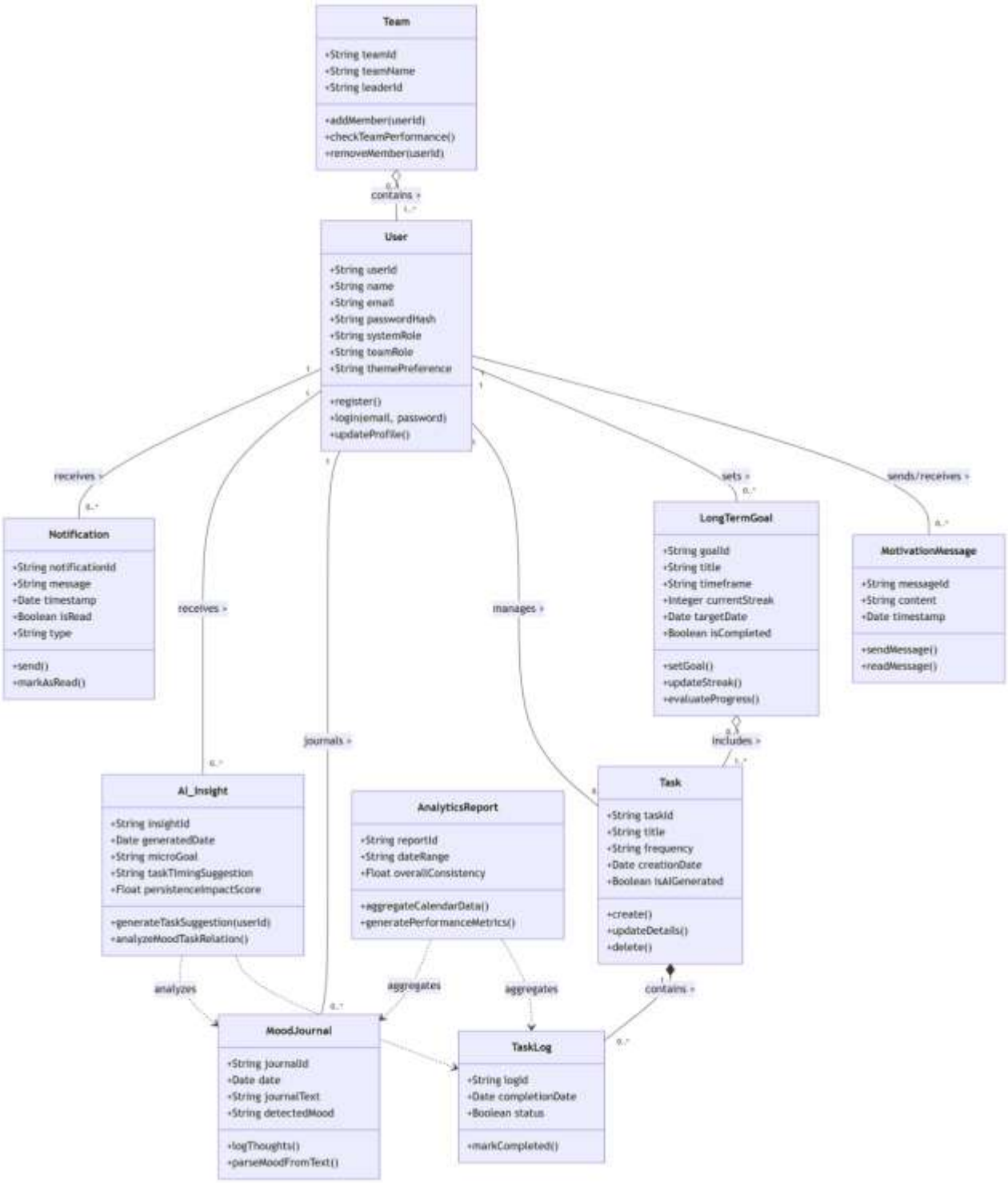
The Conceptual Model represents the pure domain logic of ConsistAI, ignoring all web-specific elements such as pages, menus, or links. It focuses entirely on the entities, their attributes, operations, and relationships.

Key Design Decisions:

- **Classes Identified:** User, Team, MotivationMessage, LongTermGoal, Task, TaskLog,

MoodJournal, AI_Insight, AnalyticsReport, and Notification.

- **Behavioral Focus:** Replaced standard "Habits" with Task and LongTermGoal to better align with the AI's generation of customized micro-goals and challenges.
- **Associations:** Established standard UML multiplicities, such as a User managing zero-to-many (0..*) Tasks, and a Composition relationship between a Task and its TaskLog.

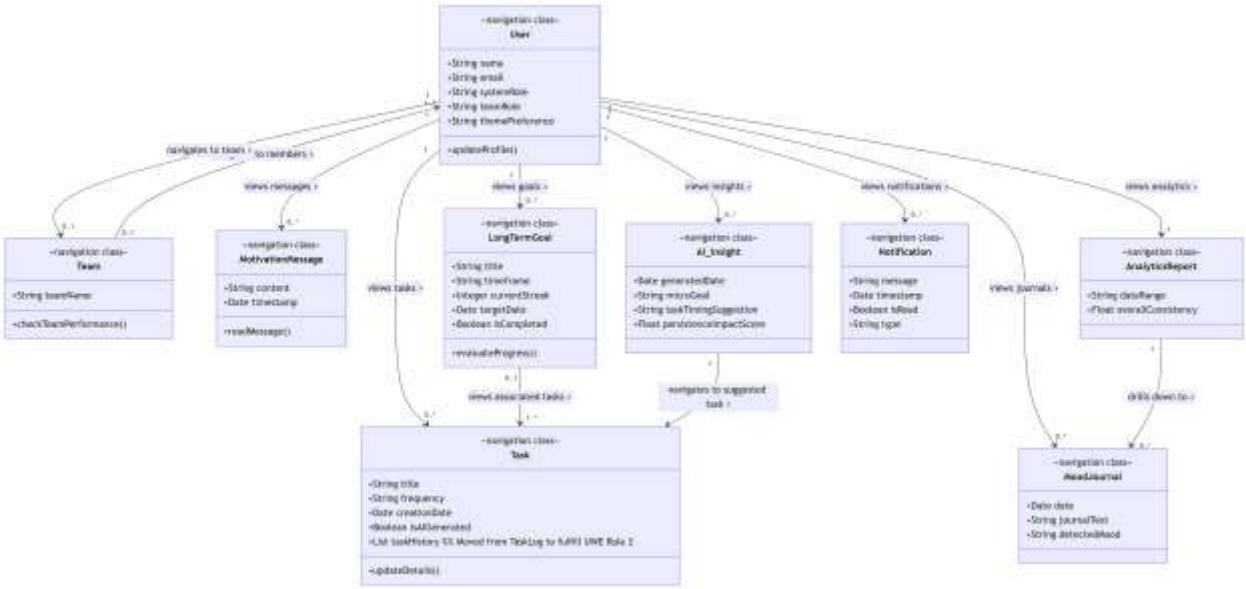


3. Phase 2: Navigational Space Model

Description:

The Navigational Space Model defines *what* information the user can navigate. It is derived directly from the Conceptual Model by applying specific UWE mapping rules.

Key Design Decisions:

- **Mapping (<<navigation class>>):** Relevant conceptual classes were stereotyped as navigation classes.
- **Omission of Non-Targets (Rule 1):** The TaskLog class was removed as it is not a standalone web page target. To preserve its data (Rule 2), it was converted into a taskHistory attribute within the Task class.
- **Presentation Relevance (Rule 2):** Backend-specific attributes such as passwordHash and all Primary/Foreign keys (userId, taskId, etc.) were removed, as they are not presented to the user on the UI.
- **Direct Navigation (Rule 3):** Directed associations were added to represent clickable paths. We also introduced "shortcuts" to avoid long navigation paths, such as a direct link from User to AnalyticsReport, and from AI_Insight to Task.
- 

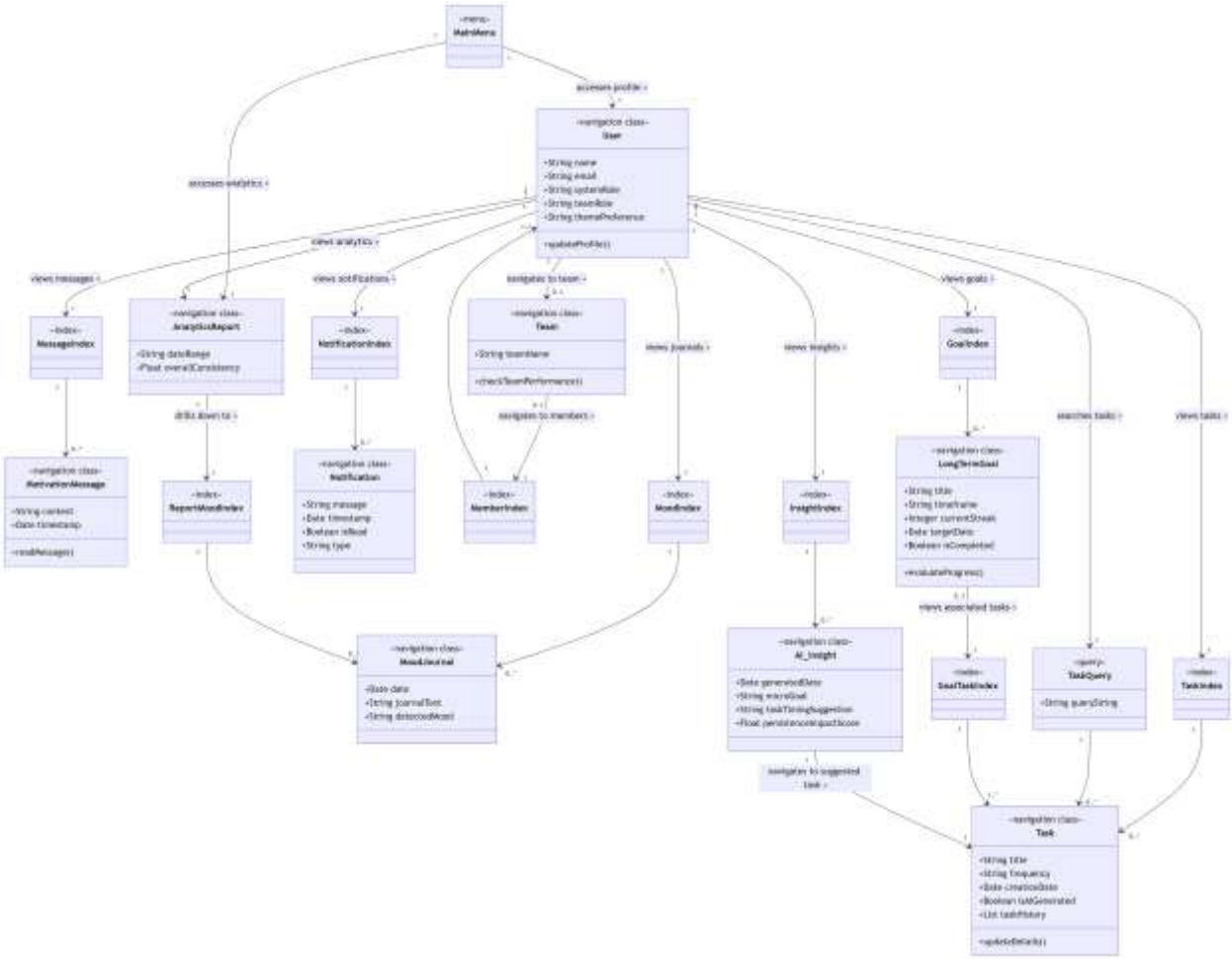
4. Phase 3: Navigational Structure Model

Description:

While the Space Model shows *what* can be visited, the Navigational Structure Model defines *how* the user reaches those targets. This is achieved by introducing **Access Elements** to mediate 1-to-many relationships.

Key Design Decisions:

- **Access Primitives:** We introduced <<index>>, <<query>>, and <<menu>> stereotypes.
- *Mediating 1-to- Links:** Any association from the Space Model targeting multiple instances (e.g., a User navigating to many Tasks) was mediated by an Access Element. For example, User -> TaskIndex -> Task.
- **Search Functionality:** We implemented a TaskQuery element to allow users to search for specific tasks, demonstrating multiple access paths to the same navigational class.
- **Role Name Shifting:** As per UWE rules, the role names (e.g., "views tasks") were shifted to the association connecting the source class to the access element.



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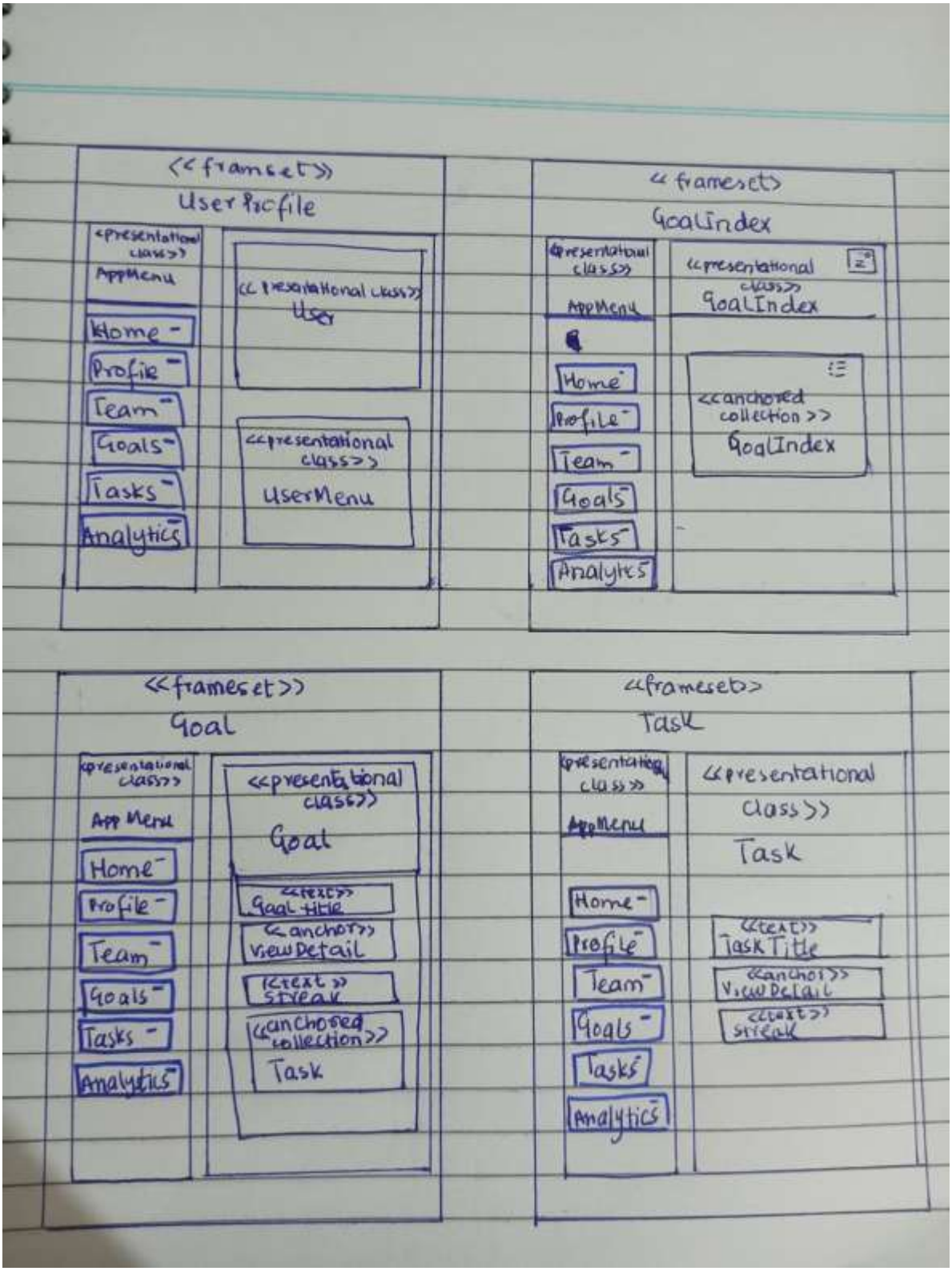
5. Phase 4: Static Presentation Model

Description:

The final phase of the assignment translates the Navigational Structure into an abstract user interface design. We utilized a **Menu-based Presentation** approach to define the structural organization of **FocusForge's** Web pages.

Key Design Decisions:

- **Presentational Classes (<<presentational class>>):** Created for every navigation class, menu, and index.
- **UI Primitives Mapping:**
  - Attributes from the navigation classes were mapped to <<text>> elements.
  - Operations (e.g., updateProfile()) were mapped to <<button>> elements.
  - Access elements like TaskIndex were mapped to <<anchored collection>>.
  - The TaskQuery was mapped to a <<form>>.
- **Frameset Composition (<<frameset>>):** We utilized framesets to model a persistent layout. Every primary view (like UserFrameset or TaskIndexFrameset) is composed of a left frame containing the MainMenuUI (which includes the mandatory "Home" <<anchor>>) and a right frame containing the specific content UI.


<<frameset>>
User Profile
<<presentational class>>
AppMenu
Home
Profile
Team
Goals
Tasks
Analytics
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User
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UserMenu
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6. Conclusion

By strictly adhering to the UWE methodology, our team successfully transitioned from a raw data-driven Conceptual Model to a highly structured, web-ready Presentation Model. This systematic approach ensures that **FocusForge's** complex features—such as AI insights, mood tracking, and social team elements—are organized in a highly navigable, user-friendly interface that avoids the "lost in hyperspace" syndrome.